

Cerebral Palsy

- Group of disorder of posture and movement that is secondary to the damage of the developing fetal or infant brain
- This is a non-progressive disorder of brain
- This contains a broad spectrum of motor disability.
- In addition to 1° impairment in gross and fine motor function, there may be problems associated with
 - Cognition
 - Seizure
 - Vision
 - Swallowing
 - Speech
 - Orthopedic deformities

Etiology :-

• Pre-natal cause

- TORCH infection
- Hypoglycemia
- Rh incompatibility
- Placental abnormalities
- Drug side effect

T - Toxoplasma

O - Others (Syphilis, HIV)

R - Rubella

C - Cytomegalovirus (CMV)

H - Herpes simplex

• Perinatal

- Pre-term baby
- ICA/MCA occlusion during birth
- Trauma
- Asphyxia
- Meningitis of neonates

- Macrogam aspiration
- low birth weight

• Post-Natal

- Delayed cry
- Jaundice
- Trauma
- Infection
- Meningitis
- Encephalitis
- Genetic

classification of CP:-

- Depending on ~~topo~~ topographical distribution

(i) Monoplegic - One limb affected

(ii) Diplegic - (LE) affected a little or no effect on (UE).

- Most common in spastic form.

- Ambulatory but scissoring gait

(iii) Hemiplegia - One side of the body

(iv) Triplegia - 3 limbs involved

(v) Quadriplegia - All limbs "

- Also k/a double hemiplegia

• Depending upon muscle tone :-

(i) Spastic CP

- Most common type of CP
- 70-80% cases
- Can be Mono, Di, Tri, Hemi or Quadri
- Cerebral cortex affected
- Generally found in antigravity muscles
- Voluntary control is always affected
- Scissoring gait ⊕⊕

(ii) Athetoid / Dyskinetic CP

- Second most common
- Mixed muscle tone
- Generally due to Bilirubin encephalopathy or Hypoxic-Ischemic Brain injury.
- Clinical Dx occurs in 18 months
- Due to injury in basal ganglia
- unwanted motor movt

(iii) Ataxic CP

- Caused due to damage of cerebellum
- Incoordination ⊕⊕
- least common type
- Voluntary movt. disymmetric.
- Generally diplegic i.e. trunk and CE affected
- Hypotonia generally
- wide BOS

(10) Mixed CP

- C/F of spastic CP and $\bar{c}$ any other type
- Generally spastic + Athetoid seen.

• Depending upon functional level

(Gross Motor Function Classification System - GMFCS)

- 5 level scale that determines a motor level for a child $\bar{c}$ motor disability.

Level I -

Level II -

Level I - walks $\bar{c}$ out limitation

Perform gross motor skill - Run, Jump etc.
Speed balance and coordination limited

Level II - walks $\bar{c}$ limitation

- Difficulty in walking long
- need assistance

Level III - walks $\bar{c}$ hand held mobility device

- wheelchair for long distance

Level IV - Power mobility

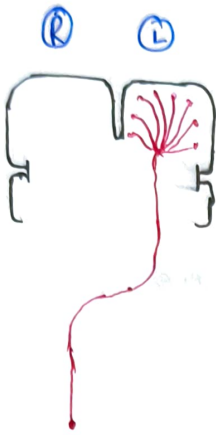
- Depend on other for wheelchair access

Level V - Transported manually

- Can't control anti-gravity muscles.

• Pathophysiology :

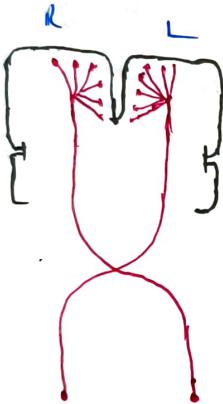
① Spastic Hemiplegic CP.



• spastic - UMN

- ↑ Deep tendon reflex
- clonus ⊕ ⊕
- ↑ Tone
- spastic
- weakness
- Babinski ⊕

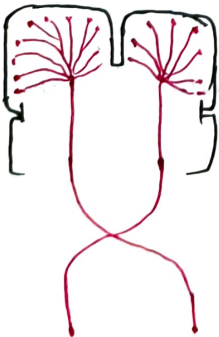
② Spastic Diplegic CP



- A/c to the Motor Homunculus, the region of cerebrum controlling the lower extremity mainly damaged.

- So, in this type mainly the LE is involved.

③ Spastic Quadriplegic CP



- All the area of both Hemunculus involved.

• Complications

- Cognitive dysfunction
- Seizures
- Vision issue
- ~~Adult~~ Auditory issue
- Due to spastic. → Scoliosis
- Speech disfn - Dysarthria
- Dysphagia
- Some times CP Patient can also have

Pseudo-Bulbar Palsy



Associated to CN - 9, 10, 12



characterised by

- Dysarthria
- Dysphagia
- Facial and Tongue weakness
- Sialorrhoea
- Aspiration pneumonia

- PT Assessment

- H/O

- Prenatal H/O

- Mother's age
- Consanguinity
- Drugs during Pregnancy
- Any infection
- H/O abortions

- Peri-natal

- Normal / Pre-term
- Labour timing
- H/O asphyxia
- Birth cry
- Type of delivery
- Child wt.
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- Post natal

- APGAR score
- Trauma
- Neonatal Meningitis / Sepsis / Hydrocephalus etc
- Nutritional habits
- Any SH, MH

- O/E

- Child alertness
- Communication
- Postural alignment

- Deformity - Kyphosis
Scoliosis
Knock knee

- usage of limbs

- Involuntary movt

- Scan

• O/E

- Orientation

- Vision

- Hearing

- Speech abnormality

- Cranial ⁽²⁾ assessment

- Sensory

- Superficial

- Deep

- Combined Cortical

- Reflex

- Spinal reflex - Flexor & dorsal reflex

Extensor thrust

Palmar and Plantar grip

Sucking and Rooting Reflex

- Tonic neck reflex

- " labyrinth "

- STNR

- ATNR

PT Management

- CP patients demonstrate impairments, functional limitations and movement dysfunction throughout their life.
- Four stages of PT is given to the CP patients from infancy to adulthood.
- The 4 stages are :-
 - ① Early intervention
 - ② Pre-School
 - ③ School age and adolescence
 - ④ Adulthood

• Goal and Aim

- The brain damage occurs in developing motor system, the 1st goal is to fasten motor development and learn functional activities

- when child learns to move 1st time



Infant's own mot provide sensory feed back



If feedback is incorrect



Mot may learned incorrectly

• Early intervention (Birth - 3 years)

- Early therapy have the impact on nervous system development and recovery from injury.
- educating the family about child's disability.

• Handling and Positioning

- " " " in supine promote head orientation in midline.
- Reaching in supine position
- Positioning of the infant in prone.
- Active head lifting in prone
- " extension " "
- Infant shouldn't sleep in prone.
- Bobath

• Feeding and Respiration

- Flexed posture facilitates feeding and social interaction b/w child and caregivers.
- Supine can be used for child to take support.

• Exercises

- Gentle ROM
- Functional activities
- Reaching
- Take proper precaution b/c sh/hip dislocation can also occur.

• CIMT

- Constraint Induced Movement Therapy.
- children b/w 18 months - 3 years.

• Second stage intervention - Pre school

- Major emphasis is to promote mobility and functional independence in the child.
- Goals during this stage are
 - ① Independent mobility
 - ② Promote functional movement
 - ③ Improve ADL
 - ④ " social interaction.

• Independent mobility

- If the child did not achieve upright and independent mobility upto a extent in early intervention, we should apply more effort in this period
- orthoses or assistive devices may be used.
- Gait training
- Muscle strengthening
- weight bearing ex.
- Trunk rotation
- Pushing a toy / Shifting weight
- Reverse walking
- Ankle foot orthosis (AFO)
 - ↓
 - Extends 10-15 mm distal to fibula head.
 - Should not pinch child behind the knee at any time

- child with unstable ankles who needs medial lateral stability → Supramalleolar orthosis (SMO)



Allows child to move freely into dorsiflexion and plantar flexion & restricting mediolateral motion

- walkers and other assistive devices

• functional movement

- Kicking balls
- Carrying objects
- Reaching overhead
- Stair climbing
- walking on ramps
- Positioning - standing, lying, sitting
- Stretching

• ADL

- learning play games
- Interacting with friends
- Always take child's level of cognitive ability into consideration when selecting a game/activity.
- Activity should be of child's interest
- Positioning

• Third stage intervention

Goals - Continue independent mobility

- Independent ADL

- Increase community / social interaction

- Increase fitness and development of positive self image.

- Studies have shown that adolescents w/ CP can ↑ strength when they are engaged in a programme of isokinetic resistance exercises.

- Strengthening improve gait and motor skills.

• Fourth stage intervention

Goals - Independent mobility and ADL

- Healthy lifestyle

- Community participation

- Independent living

- Even though the above written different goals are identified for this stage of rehab, they all are part of the role in life of an adult.

- This can be an ultimate challenge to a person w/ CP / any lifelong disability.

- Community / Social interaction should be red.

- Patient should be trained for independent ADL

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• General treatment ideas

1. child $\bar{c}$ spasticity

- Treatment focuses on mobility in all possible posture and transition b/w these postures.

- ROM exercises
- Positioning
- Active movement



Counteract the tendency to develop contractures

- Stretching

Areas prone to tightness in Quadriplegic -

- sh adductors
- elbow flexor
- wrist "
- fingers "

Areas prone to tightness in Diplegic -

- Hip flexor
- Hip adductor
- knee flexor
- Ankle plantar flexor

- weight bearing ex
- weight shifting
- slow and rhythmic rocking
- Active trunk rotation
- Active rotation of body segments
- Active resistive movement

• Advantages and Disadvantages of diff positions

① Supine

• Advantage

- Can begin early wt bearing through LE when knees bent and feet are flat on support surface
- Positioning of head and upper trunk in forward flexion can use the effect of STLR

• Disadvantage

- Supine can be disorienting because it is associated w sleeping.
- level of arousal ↓
- ~~Diff~~

② Side-lying

• Advantage

- Best position to ↓ the effect of most of the tonic reflex b/c of neutral position of Head.
- Best position to promote functional activity like rolling etc.

• Disadvantage

- Shortening of upper trunk muscles may occur if child is always positioned on same side

③ Prone

• Advantage

- Promote wt bearing on ULE (Prone on elbow/arm)
- stretch hip and knee flexors

• Disadvantage

- Breathing may be difficult
- Not recommend for young child as a sleeping posture b/c of its relationship to ↑ incidence of Sudden infant death syndrome.

④ Sitting

• Advantage

- Promotes active head and trunk control
- can provide wt bearing.
- Side sitting promotes trunk elongation / rotation
- Helpful for performing functional activity & O/E.

• Disadvantage

- child may need support on trunk
- " " unable to maintain trunk extension position due to ↓ strength or ↑ flexor tone in trunk muscles.

⑤ Quadraped

• Advantage

- wt bearing
- Trunk works anti-gravity

• Disadvantage

- flexed posture is difficult to maintain b/c of the influence of STNR.

⑥ kneeling

• Advantage

- Provide stretch to hip flexors.
 - Hip and pelvic control can be developed
- ↓
- can help in transition b/w
different postures

• Disadvantage

- Difficult to control
- child will show inability to extend at the hips completely b/c of the influence of STNR.

⑦ Standing

• Advantage

- Provide wt bearing & lower ext
- ↑ balance on low BOS

• Disadvantage

- External support may be req.